Digital Image Processing Laboratory – Lab 1

Introduction to MATLAB, Image Properties and Video Analysis

Course: Digital Image Processing

Lab No.: 1

Title: Image and Video Handling using MATLAB Online

Objective
The objective of this laboratory is to:
- Familiarize yourself with MATLAB Online and MATLAB Drive.
- Understand different image formats and image properties.
- Learn how to read and display images in MATLAB.
- Estimate image characteristics such as:
  - Image size
  - Number of bands (channels)
  - Spatial resolution
  - Quantization level (bit depth)
- Read and analyze a video file.
- Prepare a professional laboratory report.

Dataset
A folder named "Image Processing Lab 1" has been provided containing 11 sample images.
In addition, every student must include the following:
1. One selfie taken by the student.
2. One photograph of an inanimate object (e.g., bottle, laptop, chair, cup, pen, keyboard, etc.).
3. One Google image of your hometown with your home approximately at the center of the image (mark the location if necessary).

Total images to analyze = 14
- 11 provided images
- 3 student-collected images

Task 1 – MATLAB Drive
1. Log in to MATLAB Online.
2. Open MATLAB Drive.
3. Upload all 14 images into the Current Folder.
4. Also upload the provided video file.

Task 2 – Image Analysis
For **each of the 14 images**, perform the following operations.
1. Read the image
```
I = imread('filename.jpg');
```

2. Display the image

imshow(I)
title('Image')

3. Estimate image size
Display

- Number of rows
- Number of columns
- Number of bands

Hint:
size(I)

4. Determine the number of bands
Examples

- Binary image → 1 band
- Grayscale image → 1 band
- RGB image → 3 bands
- Multispectral image → Multiple bands
- Hyperspectral image → Hundreds of bands

5. Estimate quantization level (bit depth)
Determine

- Image class
- Number of bits per pixel
- Possible gray/color levels

Hints
class(I)

whos I
Interpret the results appropriately.

6. Estimate spatial resolution
Determine

- Width (pixels)
- Height (pixels)

Example
512 × 512 pixels
1920 × 1080 pixels

Record your observations
Prepare a table similar to the following.

Image No.	Filename	Size (Rows × Cols × Bands)	Image Class	Bit Depth	Quantization Levels	Spatial Resolution
1	image1.jpg	512×512×3	uint8	8-bit	256	512×512
...	...	...	...	...	...	...

Task 3 – Video Analysis

Upload the given video to MATLAB Drive.
Read the video using MATLAB.
v = VideoReader('video.mp4');
Determine the following.
Video duration
Hint
v.Duration

Frame rate (FPS)
Hint
v.FrameRate

Video width
Hint
v.Width

Video height
Hint
v.Height

Number of frames (estimate)
Hint
v.Duration * v.FrameRate

Read and display the first frame
frame1 = readFrame(v);

imshow(frame1)

Display frame size
Hint
size(frame1)

Determine
- Frame dimensions
- Number of bands
- Data type

Hints
class(frame1)

size(frame1)

Expected Learning Outcomes
After completing this laboratory, students should be able to:
- Upload files to MATLAB Drive.
- Read images and videos in MATLAB.
- Display images.

- Estimate image dimensions.
- Determine image channels (bands).
- Estimate quantization level.
- Determine image spatial resolution.
- Read and analyze a video.
- Extract and inspect a video frame.

Report Format

Your report should contain the following sections:

1. Aim
2. Software Used
3. Images Used (14 images)
4. MATLAB Commands Used
5. Image Analysis Table
6. Video Analysis
7. Screenshot(s) of MATLAB Output
8. Observations
9. Conclusion

Submission Instructions

- Prepare a single PDF report.
- Include the analysis of all 14 images and the video.
- Include relevant MATLAB code snippets and output screenshots.
- Submit the report through the designated submission portal before the start of next week's laboratory session.

Images: 14
Total Video Files: 1
Submission Deadline: Next Week's Lab